

DK1 & DK2 Energy and Ancillary Service Markets

A Reference Report for Industrial Load Market Participation

Compiled September 2026. Danish balancing-market rules evolve frequently — DK1 joined the PICASSO aFRR activation platform in October 2024, DK2 migrated FCR/FCR-N/FCR-D bidding to the Nordic MMS platform in October 2025, and minimum bid sizes have been progressively lowered over recent years. Always confirm current parameters against Energinet's live market pages before submitting bids.

1. Energy Markets (DK1 and DK2)

Market	Min. Bid / Resolution	Opening / Closure	Symmetric?	Notes	What it means for an industrial unit
Day-ahead (Nord Pool spot)	No formal minimum; ~0.1 MWh typical granularity	Opens ~2 days ahead; closes 12:00 CET day-ahead	N/A — separate buy/sell bids	Sets the reference price for the delivery day; consumption is scheduled here.	Base tool for load-shifting: move flexible production to cheap hours, avoid expensive ones. Standard supplier/BRP trade — no prequalification needed.
Intraday (Nord Pool continuous)	0.1 MW, 0.1 MW resolution	Opens after day-ahead results (~14:00–15:00); trades until ~5–60 min before delivery (border-dependent)	N/A	Corrects the day-ahead position as forecasts or plans change.	Useful when flexibility only becomes clear close to real time (e.g. an unplanned batch delay) — re-trade your position instead of carrying full imbalance risk.
Regulating power / mFRR energy activation (imbalance settlement)	Same as mFRR capacity bid; activated in 15-min quarters, 1 MW steps	Bids updatable to ~45 min before delivery; gate closure ~21:30 D-1 for standing bids	Separate up/down bids	The real-time balancing layer; non-participants are still settled against the imbalance price.	Even without formal ancillary-service bids, deviations from your schedule are settled at the imbalance price — a key reason to actively manage flexible load.

2. Ancillary (Balancing) Service Markets — DK1

DK1 (western Denmark) is part of the Continental Europe synchronous area.

Market	Min. Bid / Resolution	Opening / Closure	Symmetric?	Notes	What it means for an industrial unit
FCR (FCR Cooperation — 9-country market: DK, DE, AT, BE, CH, FR, NL, CZ, SI)	1 MW, resolution 1 MW; indivisible bids up to 25 MW	Gate opens D-1 ~11:00; closes D-1 08:00 CET; 4-hour blocks, 6 auctions/day	Symmetric — equal MW up and down	Fastest product (response ≤30s); capacity-only payment, no separate energy settlement; DK must source ≥30% locally; min. bid lowered over time from 10 MW → 5 MW → 1 MW.	Most accessible entry point for a mid-size industrial load. Commit to ±1 MW automatic response to frequency; symmetric design suits loads with roughly balanced up/down flexibility (compressors, cooling, e-boilers). Revenue is mostly the capacity payment since delivered energy nets out.
aFRR — capacity market	5 MW	Day-ahead auction (Nordic capacity market framework)	Separate up/down products	Higher bar than FCR; product design still favours larger or aggregated portfolios.	A single mid-size plant usually can't reach 5 MW alone — pooling several sites via an aggregator/BSP becomes necessary. Best suited to larger flexible processes or portfolios.
aFRR — energy activation (PICASSO)	Linked to capacity bid	Real-time, centrally dispatched by Energinet	Separate up/down	DK1 joined PICASSO with DE, AT, CZ on 2 Oct 2024; prices have spiked above €4,000/MWh during scarcity events.	Clearing the capacity market obligates you to also bid activation and be centrally dispatched — less 'opt-in' than FCR. Understand this automatic-dispatch obligation before committing capacity.
mFRR — capacity market	1 MW, 1 MW steps (trilateral DK–SE–FI market)	Capacity market closes 09:30 D-1 (before day-ahead spot closes); activation price bids due 17:00 D-1	Asymmetric — DK1 procures upward regulation only	~600 MW procured per hour in DK1; open borders with SE/FI mean cross-border bids can clear local needs.	Naturally suited to load shedding — if your plant can curtail on a manual signal within 15 minutes, this is one of the more approachable products (low 1 MW threshold, no frequency-following hardware needed). No downward product exists in DK1, so only demand reduction is rewarded.
mFRR — energy activation	Linked to capacity bid (or voluntary bids)	Merit-order activation in real time; updatable to ~45 min before delivery	Up only in DK1	Paid the marginal activation price when actually called; capacity-cleared bids are obligated to also offer here.	Two revenue legs: a guaranteed capacity payment (called or not) plus an activation payment only when actually dispatched. Suits processes tolerant of occasional, unpredictable short curtailments.
Reactive power / voltage support	Plant-specific	Bilateral/contractual — not a liquid market	N/A	Primarily relevant to large generators and grid-connected converters.	Generally not accessible to a typical industrial consumption load — included for completeness only.

3. Ancillary (Balancing) Service Markets — DK2

DK2 (eastern Denmark) is part of the Nordic synchronous area and needs a richer product set than DK1 because it sits on the smaller Nordic system rather than the robust Continental European grid.

Market	Min. Bid / Resolution	Opening / Closure	Symmetric?	Notes	What it means for an industrial unit
FCR-N (normal operation)	Small (sub-MW territory) — confirm current value on the Nordic MMS platform (DK2 migrated Oct 2025)	Two auctions: D-2 (~80% of volume) and D-1 (remainder), both pay-as-bid	Symmetric	Activated continuously for small deviations, 49.9–50.1 Hz band.	Constant, low-amplitude regulation duty — suited to assets making small, continuous, bidirectional adjustments (e.g. fine-controlled HVAC/refrigeration), not simple on/off loads.
FCR-D (disturbance — dynamic and static sub-products)	~0.1 MW minimum bid; bid durations flexible, 1–6 hours	Same D-2 / D-1 auction structure as FCR-N	Asymmetric — separate FCR-D up and FCR-D down products	Fast, large-amplitude response to bigger frequency events. 'Static' variant gives up to a 15-min grace period for entities that can't meet the stricter 'dynamic' ramp/stability requirements. Pooled bids capped around 3 MW per pool, adding metering/admin overhead.	The most accessible DK2 product for smaller industrial loads: very low minimum bid, directional (not symmetric) commitment so you can offer only the direction your process supports (e.g. up-regulation via shedding), and the 'static' option suits slower-responding equipment.
FFR (Fast Frequency Reserve)	Small — comparable order to FCR-D; confirm via Energinet portal	Closes after the day-ahead market, day before operation	Directional (up); combinable bid option with FCR-D	Newer product (since 2020) for very fast (<1s) response to large disturbances (e.g. loss of a major generator/interconnector).	Relevant mainly for loads with genuine sub-second disconnection capability (e.g. protection-relay-triggered shedding) — a narrower niche than FCR-D for most industrial processes.
aFRR — capacity market	5 MW (Nordic aFRR capacity market)	Day-ahead Nordic auction	Separate up/down	Same Nordic capacity market DK1 now also participates in.	Same aggregation logic as DK1 — a single mid-size plant usually needs pooling to reach 5 MW.
aFRR — energy activation (PICASSO)	Linked to capacity	Real-time	Separate up/down	DK2 was already in the Nordic aFRR capacity market since Dec 2022, ahead of DK1.	Same automatic-dispatch obligation as DK1 once capacity clears.
mFRR — capacity market	Confirm current Nordic MMS value (trilateral DK–SE–FI structure; historically 1 MW in the common market)	Similar structure to DK1; DK2 also runs a monthly auction (~360 MW) alongside hourly (~240 MW)	Symmetric availability — DK2 procures both upward and downward mFRR capacity	More balancing depth than DK1 because DK2 sits in the smaller, less self-sufficient Nordic system.	Because DK2 buys downward mFRR too, a load here that can increase consumption on signal (not just shed it) has a real revenue opportunity that doesn't exist in DK1 — relevant for batch electrolysis, heat pumps, e-boilers.
mFRR — energy activation	Linked to capacity bid	Merit-order, real-time	Up and down	Same mechanics as DK1 but bidirectional.	Same two-part revenue logic (capacity + activation), plus an additional downward-regulation revenue stream unavailable in DK1.

4. Quick Decision Guide for an Industrial Unit

If your load can...	Best-fit markets
Only shift consumption to cheaper hours (predictable, day-ahead planning)	Day-ahead market
React to short-notice forecast changes	Intraday market
Shed load automatically & continuously in proportion to frequency deviations, roughly symmetric	FCR (DK1) / FCR-N (DK2)
Shed a fixed block of load on a fast trigger, direction-only (up), with no need to be compensated for downward flex	FCR-D (DK2), or mFRR (DK1)
Increase consumption on signal (not just cut it)	mFRR down or aFRR down — DK2 only; not available in DK1
Trip within <1 second on major disturbances	FFR (DK2)
Only reach flexible capacity below ~1–5 MW alone	Start with FCR / FCR-D (lowest minimum bids); use an aggregator/BSP for aFRR or larger mFRR positions
Reach 5 MW+ of flexible capacity alone	Direct aFRR participation becomes realistic without pooling

5. Key Sources to Verify Live Parameters Before Bidding

- Energinet English ancillary services portal — en.energinet.dk/electricity/ancillary-services/
- Energinet prequalification documents (technical test requirements per product)
- Energinet “Outlook for Ancillary Services 2024–2040” (volume/price projections, market development roadmap)
- ENTSO-E FCR Cooperation page (continental FCR rules, DK1's role)
- Energi Data Service (Energinet open data portal) for historical and live prices/volumes across all products

Note: some figures above (exact FCR-N/FCR-D minimum bid on the new Nordic MMS platform, current mFRR trilateral minimum bid) are in a state of active change following DK2's October 2025 platform migration — confirm current values directly with Energinet before finalising a bidding strategy.